

Module 02 – Linux Fundamentals (Basic Commands & Introduction)

1. Operating System (OS)

- An Operating System acts as an interface between the user and computer hardware.
 - It manages files, memory, processes, input/output devices, and system resources.
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2. What is Linux?

- Linux is an open-source operating system based on the Linux Kernel.
 - It was created by **Linus Torvalds** in **1991**.
 - The kernel manages communication between hardware and software.
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3. Why Learn Linux?

- Widely used by IT professionals and cybersecurity experts.
 - Open source and customizable.
 - Lightweight, secure, and reliable.
 - Commonly used for servers and ethical hacking.
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4. Where is Linux Used?

- Web servers
 - Cloud computing
 - Embedded systems
 - Traffic control systems
 - Industrial automation
 - Point of Sale (POS) systems
 - Car infotainment systems
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5. Linux Architecture

Linux consists of:

- **Kernel** – Controls hardware and system resources.
- **Shell** – Accepts user commands.
- **System Libraries** – Provide reusable functions.

Module 02 – Linux Fundamentals (Basic Commands & Introduction)

- **System Utilities** – Perform specific system tasks.
 - **Hardware Layer** – CPU, RAM, Storage, etc.
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6. Linux Flavours (Distributions)

Linux has multiple distributions, including:

- Ubuntu
- Debian
- Fedora
- Kali Linux
- CentOS
- Arch Linux

Each distribution is designed for different purposes.

7. Windows vs Linux

Windows

- Uses drive letters (C:, D:, E:)
- Proprietary operating system
- GUI-focused

Linux

- Uses a single root (/) directory
 - Open source
 - Strong command-line interface
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8. Linux File Hierarchy

Linux organizes everything under the root directory (/).

Common directories:

- `/bin` – Essential commands
- `/boot` – Boot files

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- /dev – Device files
 - /etc – Configuration files
 - /home – User directories
 - /lib – Libraries
 - /media – Removable devices
 - /mnt – Mounted file systems
 - /opt – Optional software
 - /root – Root user files
 - /tmp – Temporary files
 - /usr – User programs
 - /var – Variable data and logs
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9. Basic Linux Commands

- pwd – Display current directory.
 - ls – List files and folders.
 - cd – Change directory.
 - mkdir – Create a new directory.
 - rmdir – Remove an empty directory.
 - touch – Create a new file.
 - cp – Copy files or directories.
 - mv – Move or rename files.
 - rm – Delete files.
 - cat – Display file contents.
 - clear – Clear the terminal screen.
 - history – Show previously executed commands.
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10. File Permissions

Linux controls access using:

- Read (r)
- Write (w)
- Execute (x)

Permissions apply to:

- Owner
- Group
- Others

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11. Text Editors

Common Linux text editors:

- Nano
- Vim
- Vi

Nano is beginner-friendly and widely used.

12. Package Management

Install software using package managers.

Examples:

```
sudo apt update
sudo apt install <package-name>
```

13. Process Management

Useful commands:

- `ps` – View running processes.
 - `top` – Monitor system processes.
 - `kill` – Terminate a process.
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14. SSH

SSH (Secure Shell) allows secure remote access to another system using an encrypted connection.

Module 02 – Linux Fundamentals (Basic Commands & Introduction)

15. System Monitoring

Commands used to monitor the system:

- `top`
- `htop`
- `df`
- `free`

These help check CPU, memory, and disk usage.

16. Log Management

Linux stores system logs to help monitor activity and troubleshoot issues.

Key Takeaways

- Understood the role of Linux as an operating system.
- Learned Linux architecture and file hierarchy.
- Practiced essential Linux commands.
- Explored file permissions, package management, process management, SSH, and system monitoring.
- Built a strong foundation for working with Linux in cybersecurity.